



Chapter 1

Introduction

Partially Edited and
Presented by
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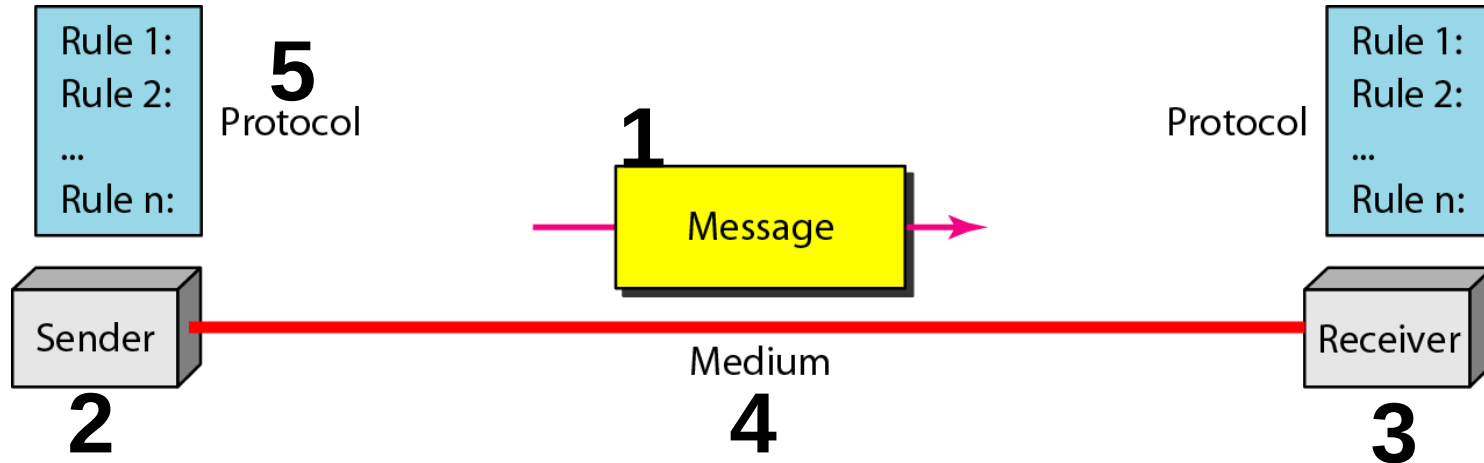
1-1 DATA COMMUNICATIONS

- ❖ *The term **telecommunication** means communication at a distance.*
- ❖ *The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data.*
- ❖ ***Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable or in wireless form. Major Characteristics:*
 - ❖ *Delivery-System must deliver data to the correct destination*
 - ❖ *Accuracy-System must deliver data accurately*
 - ❖ *Timeliness-System must deliver data in a timely manner*
 - ❖ ***Jitter**- minimize the jitter of the delivery*

Components- Data Communication

- 1. Message.** The **message** is the information (data) to be communicated between sender and receiver. Popular forms of information include text, numbers, pictures, audio, and video.
- 2. Sender.** The **sender** is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
- 3. Receiver.** The **receiver** is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
- 4. Transmission medium.** The **transmission medium** is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
- 5. Protocol.** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Figure 1.1 *Five components of data communication*



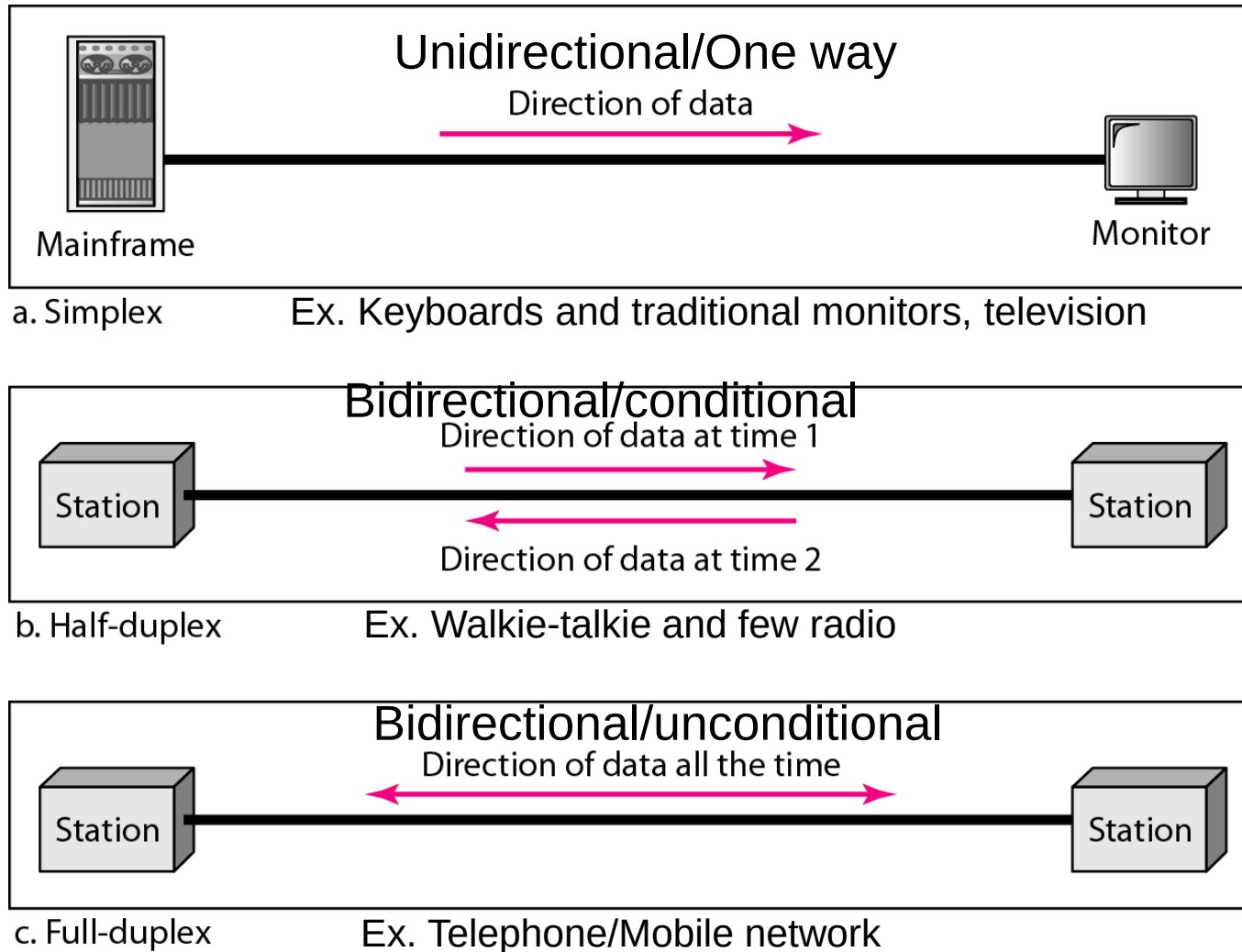
Data Representation

- Text (Unicode, American Standard Code for Information Interchange (ASCII))
- Numbers
- Images (Divided into pixel with size in bit patterns, has color scheme of RGB- *red*, *green*, and *blue* or YCM- *yellow*, *cyan*, and *magenta*.)
- Audio
- Video

Data Flow

- Simplex
- Duplex
 - Half Duplex
 - Full Duplex

Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



1-2 NETWORKS

- A **network** is a set of devices (often referred to as **nodes**) connected by communication **links**.
- A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

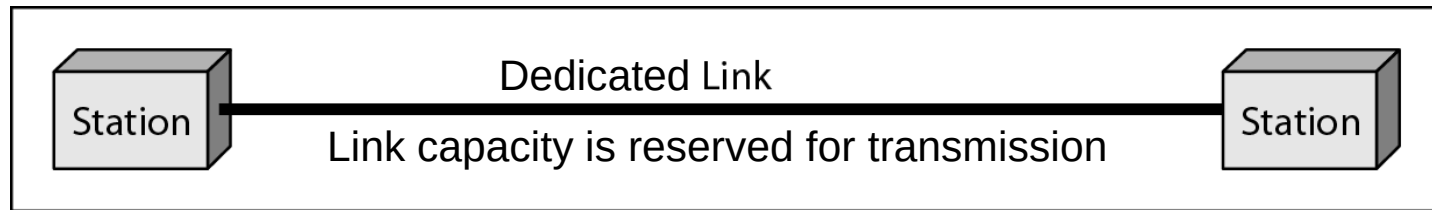
Topics discussed in this section:

- Network Criteria (performance(throughput and delay), reliability, security)
- Physical Structures (Type of Connection: Multipoint, Point to Point.
- Topology: Mesh, Star, Bus, Ring, Hybrid)
- Network Models (OSI, TCP/IP)
- Categories of Networks (PAN, LAN, MAN, WAN)
- Interconnection of Networks: Internetwork

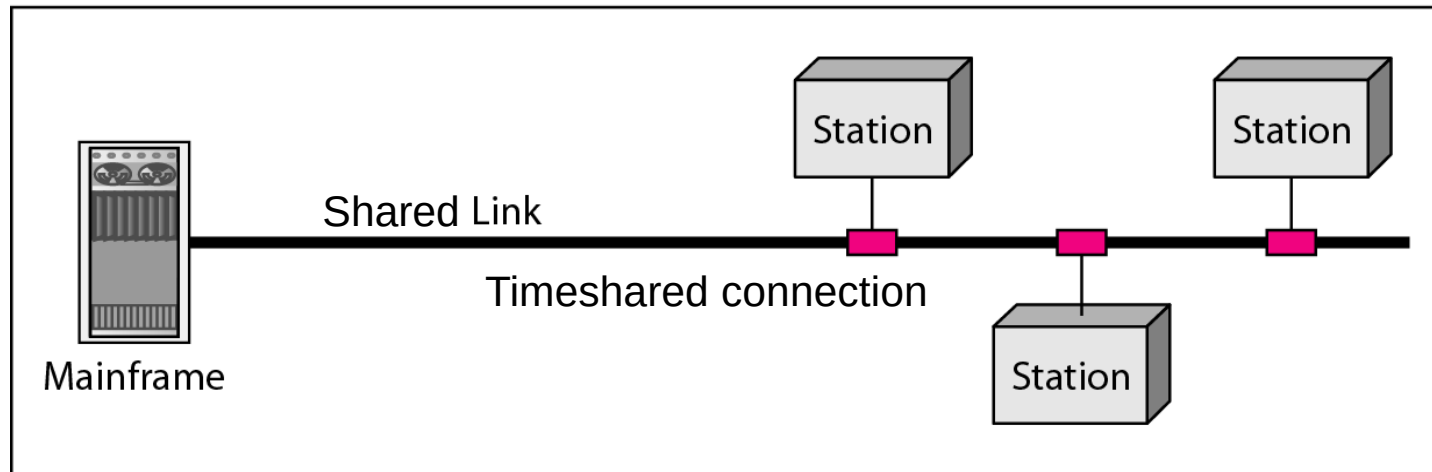
Network Criteria

- Performance can be measured in many ways as **Transit** and **Response** Time and **Throughput** and **Delay**.
- **Transit time** is the amount of time required for a message to travel from one device to another.
- **Response time** is the elapsed time between an inquiry and a response.
- **Throughput** is the amount of data successfully received from a device.
- **Delay** defines the total time of an entire message arrived at the destination after the first bit of the message sent from the source device

Figure 1.3 *Types of connections: point-to-point and multipoint*



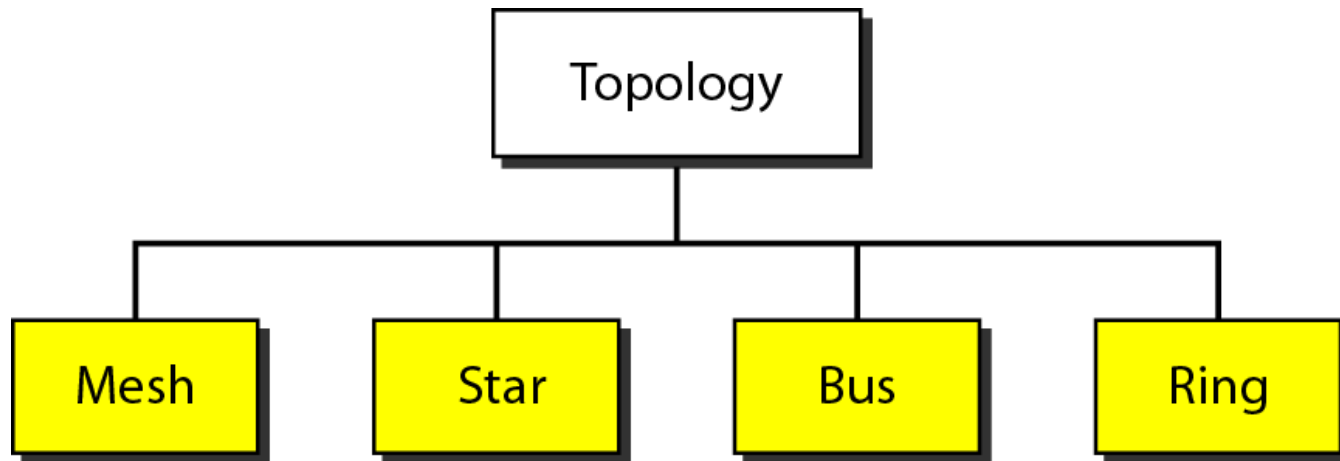
a. Point-to-point Ex. Microwave or satellite links

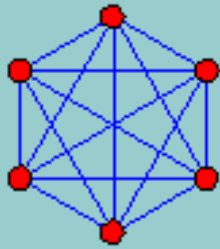


b. Multipoint Ex. Bus topology

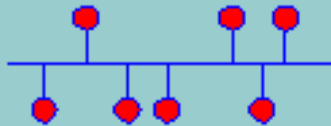
Figure 1.4 *Network topology*

Network topology is the **geometric** representation of all the links and linking devices(usually called **nodes**) to one another.

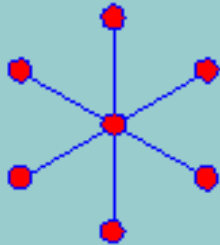




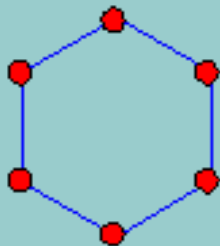
a) Fully Connected Topology



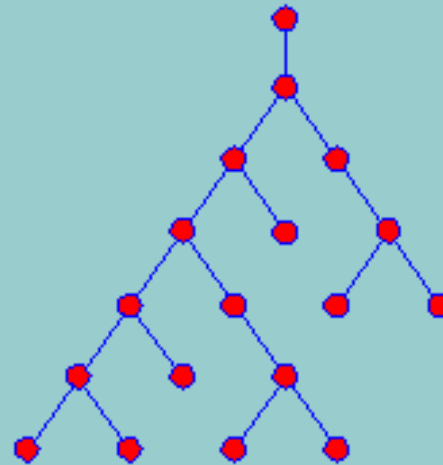
b) Bus Topology



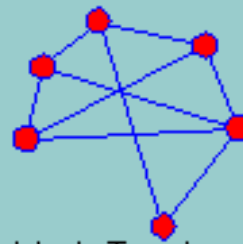
d) Star Topology



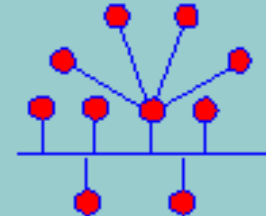
d) Ring Topology



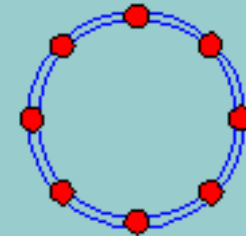
e) Tree Topology



f) Mesh Topology



g) Hybrid Topology
(example: combination of
Star topology and Bus topology)



h) Dual Ring Topology



i) Linear Topology

Nodes ● — Branches

Figure 1.5 *A fully connected mesh topology (five devices)*

- Dedicated point-to-point link
- If n nodes exist, then each node connected $n-1$ nodes or link.
- So, for n nodes need to connected total $n(n-1)$ physical links
- For duplex mode, divide the total number of links by 2 means

$$\frac{n(n-1)}{2}$$

Duplex mode links present in a mesh network

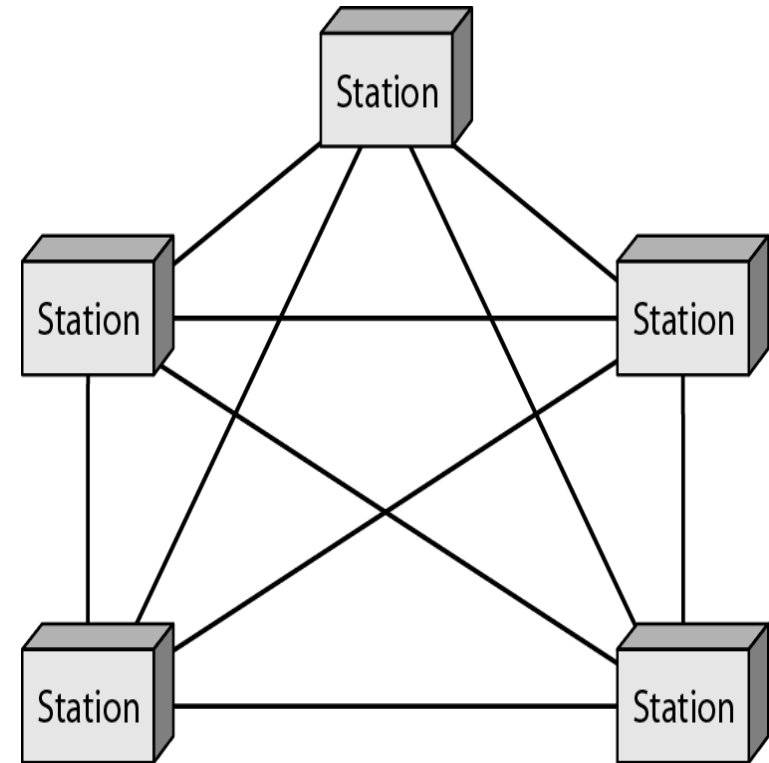


Fig. Mesh Topology

Figure 1.6 *A star topology connecting four stations*

- Dedicated point-to-point link to a central hub
- No direct link between nodes
- Hub or controller acts as media to send data to one another
- Each node connected to the controller via a single (one) links
- If only one link fails, only that links affected.
- This topology used in LAN communication.
- For n number of stations, there are n links to connect.

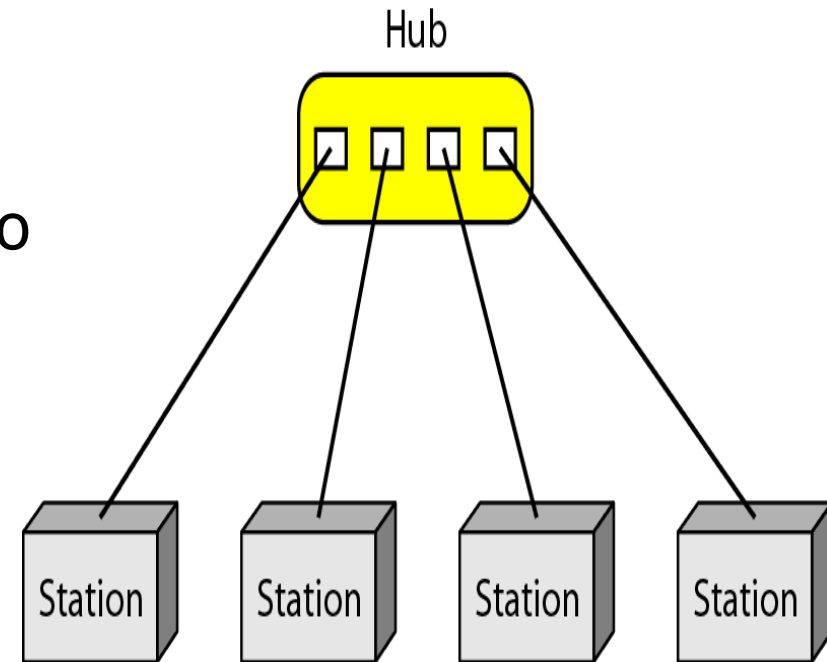
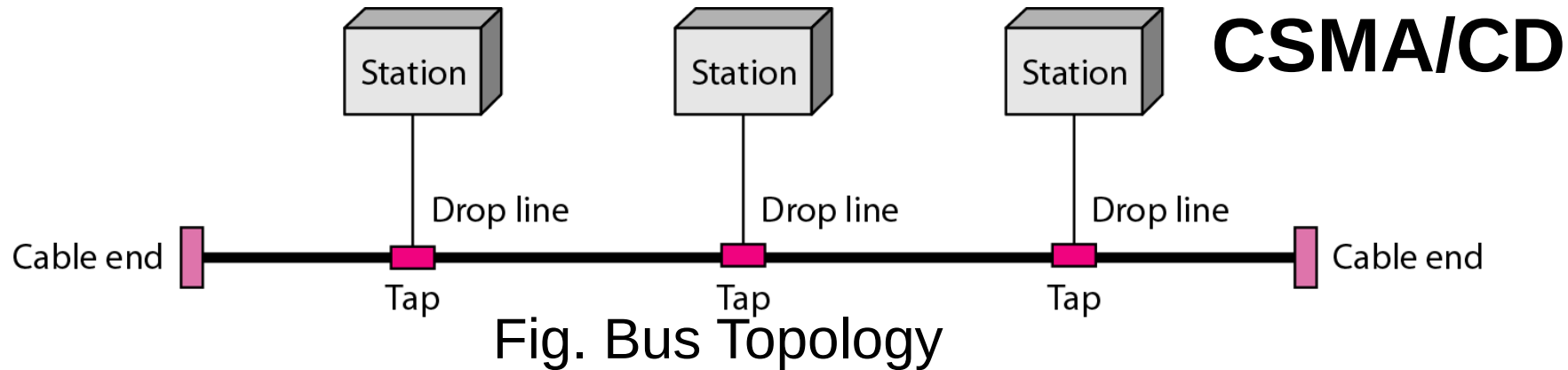


Fig. Star Topology

CSMA/CD, CSMA/CA

Figure 1.7 *A bus topology connecting three stations*



- Single connection is shared(multipoint) by multiple nodes
- Signal becomes weak as it travels further distance
- Signal reflection at tapes reduces the signal quality.
- A fault or break in the bus cable stops all transmission.
- Advantages:
 - Ease of Installation.
 - Uses less cables.
 - Less complex.
- Disadvantages:
 - Difficult to fault isolation and reconnection.
 - Difficult to add new devices.

Figure 1.8 *A ring topology connecting six stations*

- Dedicated point-to-point connection with two neighbor devices on each side
- Signal passed along the ring in one direction, until it reaches the destination.
- Each device has a repeater to regenerate the bits
- If the signal is not intended to a receiver that repeater regenerate the signal and passes through the ring.
- Advantages:
 - Easy to install and reconfigure
 - Fault isolation is simplified
- Disadvantages:
 - Unidirectional traffic(maximum ring length and number of devices)
 - A break in the ring (such as a disabled station) can disable the entire network. Solution: Dual Ring.

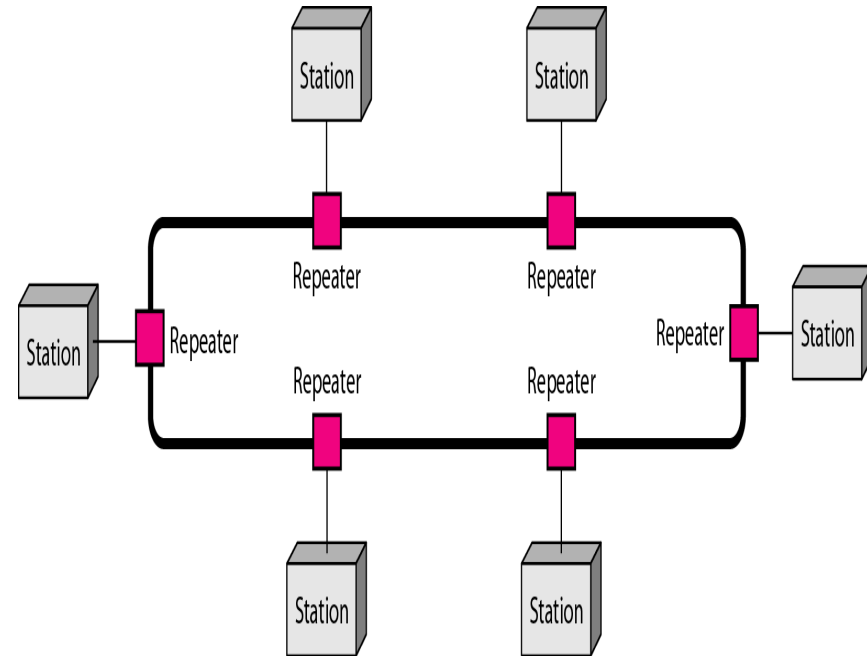
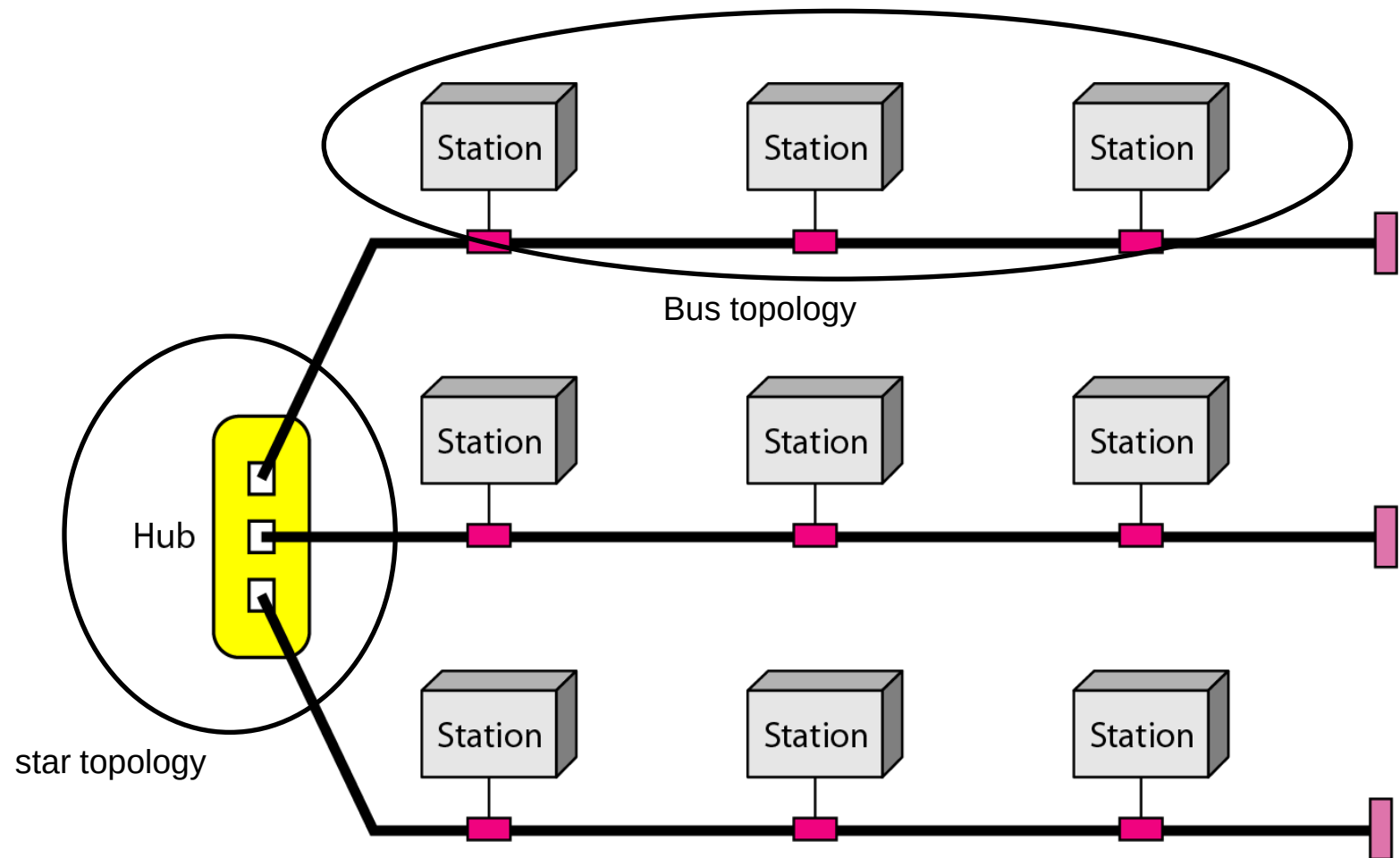


Fig. Ring Topology
Token Ring protocol

Figure 1.9 *A hybrid topology: a star backbone with three bus networks*



A hybrid topology

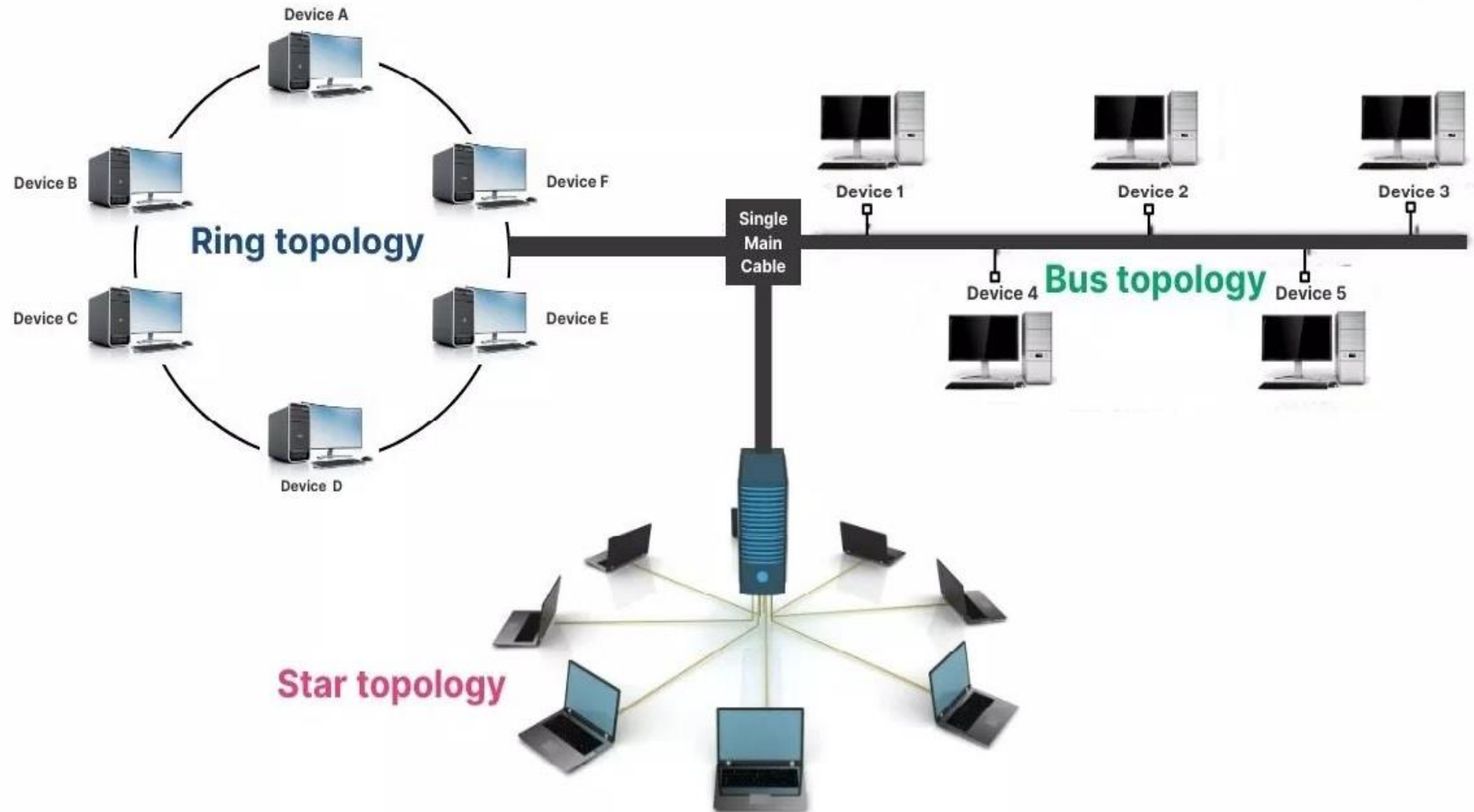
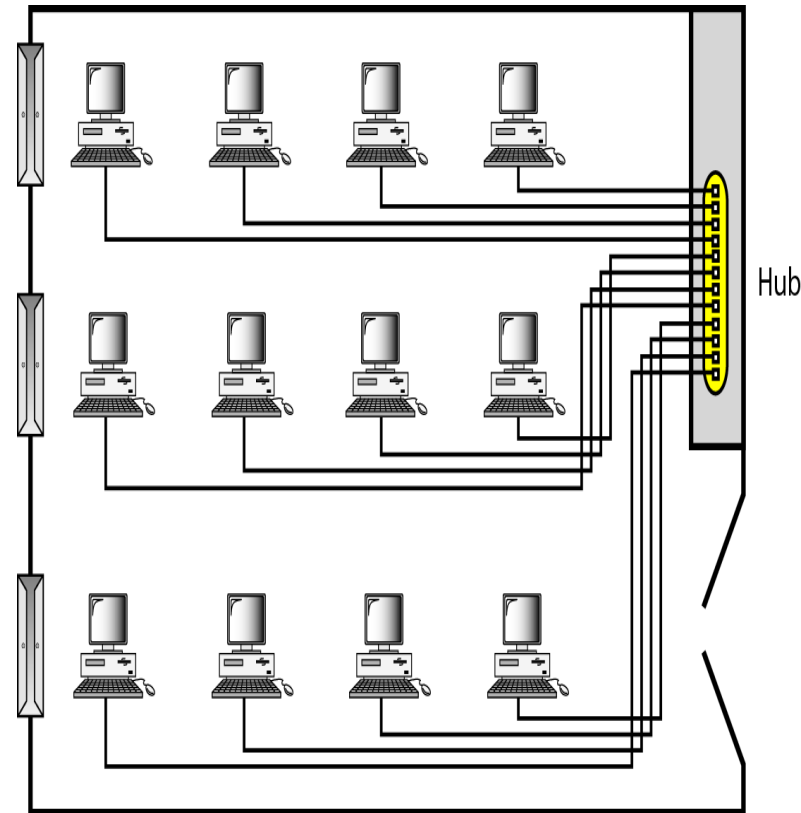


Figure 1.10 *An isolated LAN connecting 12 computers to a hub in a closet*

- Local Area Network(LAN) – Privately owned network and links a single office, building, or campus.
- LANs are designed to share resources among the computers as printers, scanner
- Some LANs connect workstations in a group called domain controller.
- Currently, LAN size is limited to a few kilometers.
- Early LANs had data rates in 4 to 16 Mbps but currently it is 100 or 1000 Mbps



Metropolitan Area Network(MAN)

- Large network that connects different organizations
- It might connect different corporate LANs together
- Controlled or own by multiple organization
- Maintain by a group or single network provider(Link3)
- Share regional resources.
- Apex or different showroom`s are interconnected
- A telephone network which share high speed DSL distributed to customer also a good example of MAN

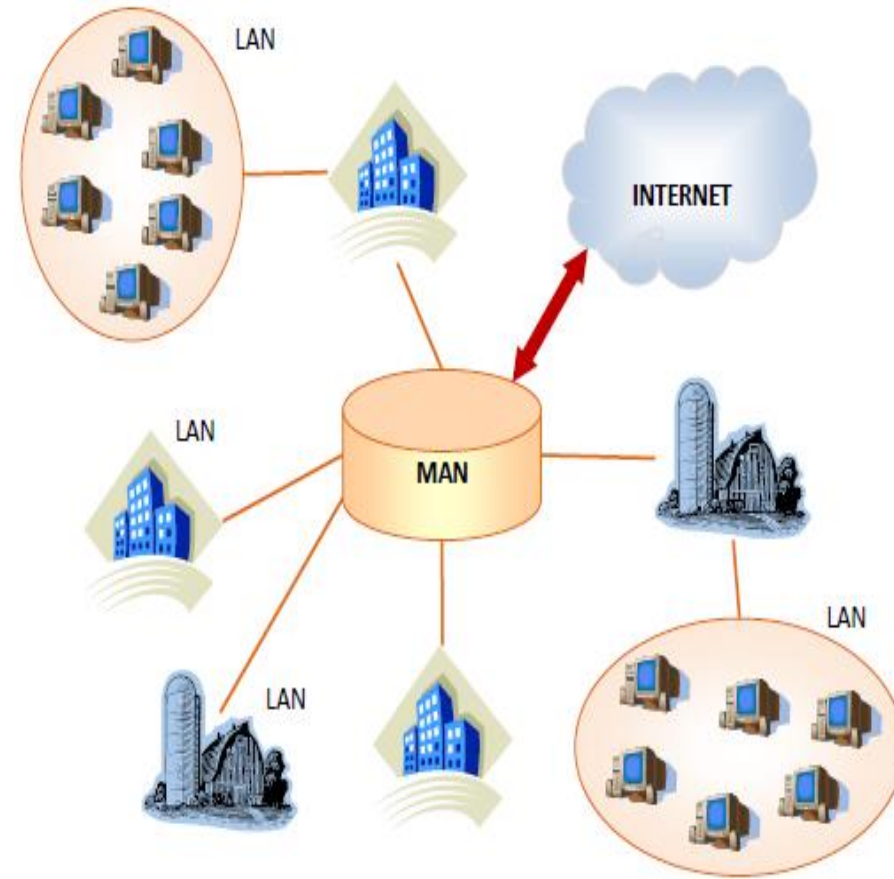
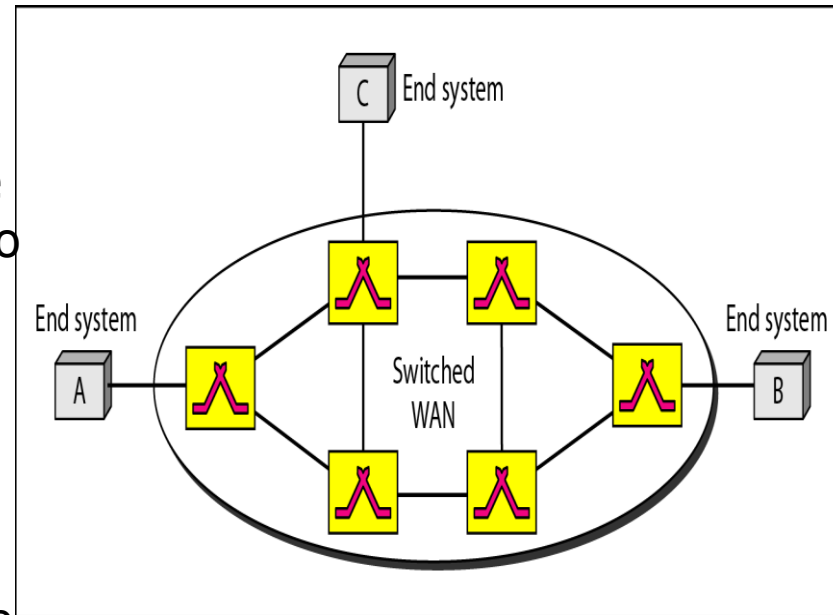
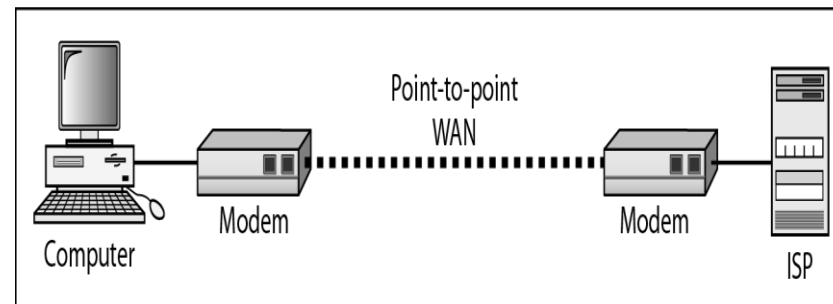


Figure 1.11 *WANs: a switched WAN and a point-to-point WAN*

- Wide Area Network(LAN) – covers large geographical area
- WANs are designed to provide long distance transmission of data, image, audio and video information
- A WAN can be
 - Complex backbone network that connects the internet called switched WAN or
 - Simple dial up line that connects a home computer to the internet called point-to-point WAN
- Early example of switched WAN is X.25 and also current ATM service example of WAN service.



a. Switched WAN



b. Point-to-point WAN

1-3 THE INTERNET

- ❖ The **Internet** has revolutionized many aspects of our daily lives.
- ❖ It has affected the way we do business as well as the way we spend our leisure time.
- ❖ The Internet is a communication system that has brought a wealth of information to our fingertips and organized it for our use.

Topics discussed in this section:

A Brief History

The Internet Today (ISPs)

Brief History

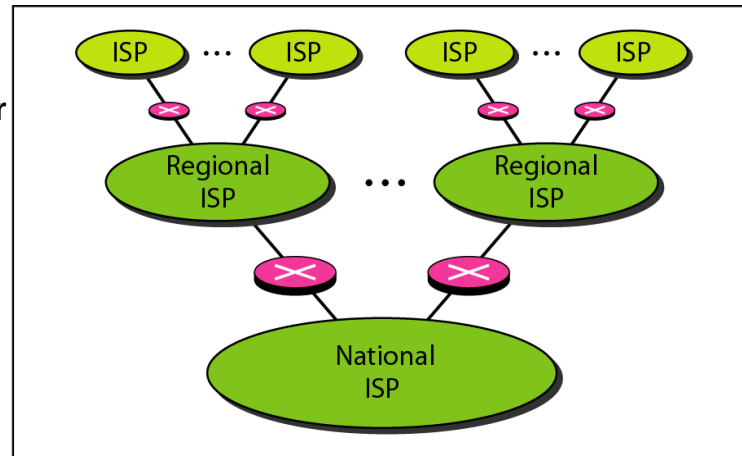
- Came into existence in 1969 only after Packet Switched Network was invented in 1961 at MIT by Leonard Kleinrock.
- DoD (Dept of Defense) through ARPA (Advanced Research Projects Agency) represented ARPANET in ACM (Association of Computing Machinery) meeting in 1967.
- Introduced IMP (Interface Message Processor).
- In 1969, UCLA – UCSB – SRI – UoU got connected.
 - University of California Los Angeles – UCLA
 - University of California Santa Barbara – UCSB
 - University of Utah – UoU
 - Sanford Research Institute (SRI)
- Software Support by NCP – Network Control Protocol.
- 1972, Vint Cerf and Bob Kahn came up with paper on TCP with networking concepts of encapsulation, datagram, gateways, end to end delivery of packets.
- Soon TCP was split into TCP and IP.
- In 1981, UC Berkeley modified the UNIX operating system to include TCP/IP which popularized Internetworking.
- Then came creation of CSNET in 1981. **Computer Science Network (CSNET) was a network sponsored by the National Science Foundation (NSF).**
- In 1983, ARPANET split into two networks: **Military Network (MILNET) for military users** and ARPANET for nonmilitary users.

1.23 ■ Today, it is managed by ISP – Internet Service Providers.

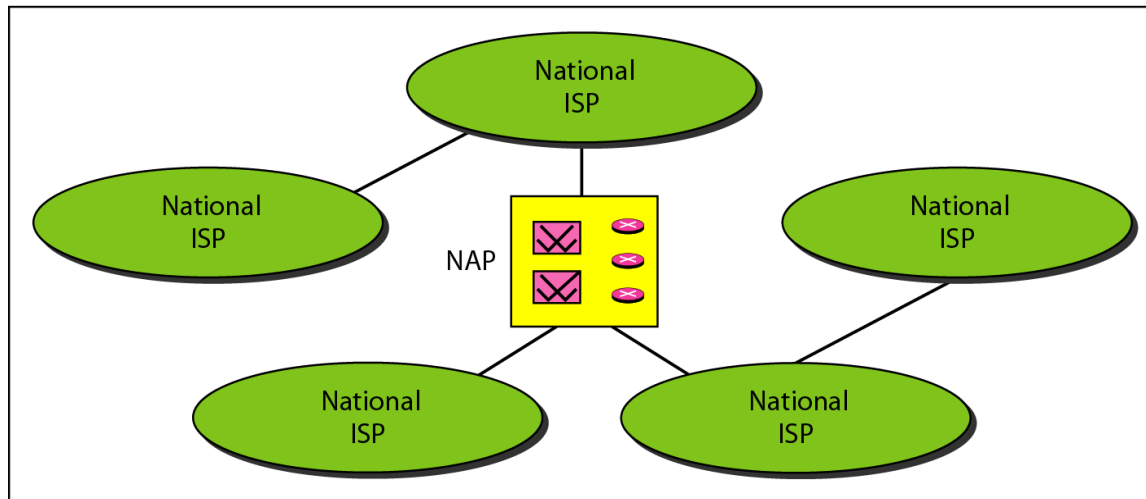
Figure 1.13 *Hierarchical organization of the Internet*

ISP – Internet Service Provider
NAP – Network Access Point

ISP Communicate with each
others network through NAP



a. Structure of a national ISP



b. Interconnection of national ISPs

Standards Organization

- ISO – International Organization of Standardization
- ITU-T – International Telecommunication Union
Telecommunication Standards formed CCITT in 1993 for research.
- CCITT – Consultative Committee for International Telegraphy and Telephony.
- ANSI – American National Standards Institutions.
- IEEE – Institute of Electric and Electronics Engineers
- EIA – Electronic Industries Association
- FCC – Federal Communication Commission
- RFC – Request for Comment
- How about Controlling body in India?

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number. Nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth)
802.16 *	Broadband wireless
802.17	Resilient packet ring